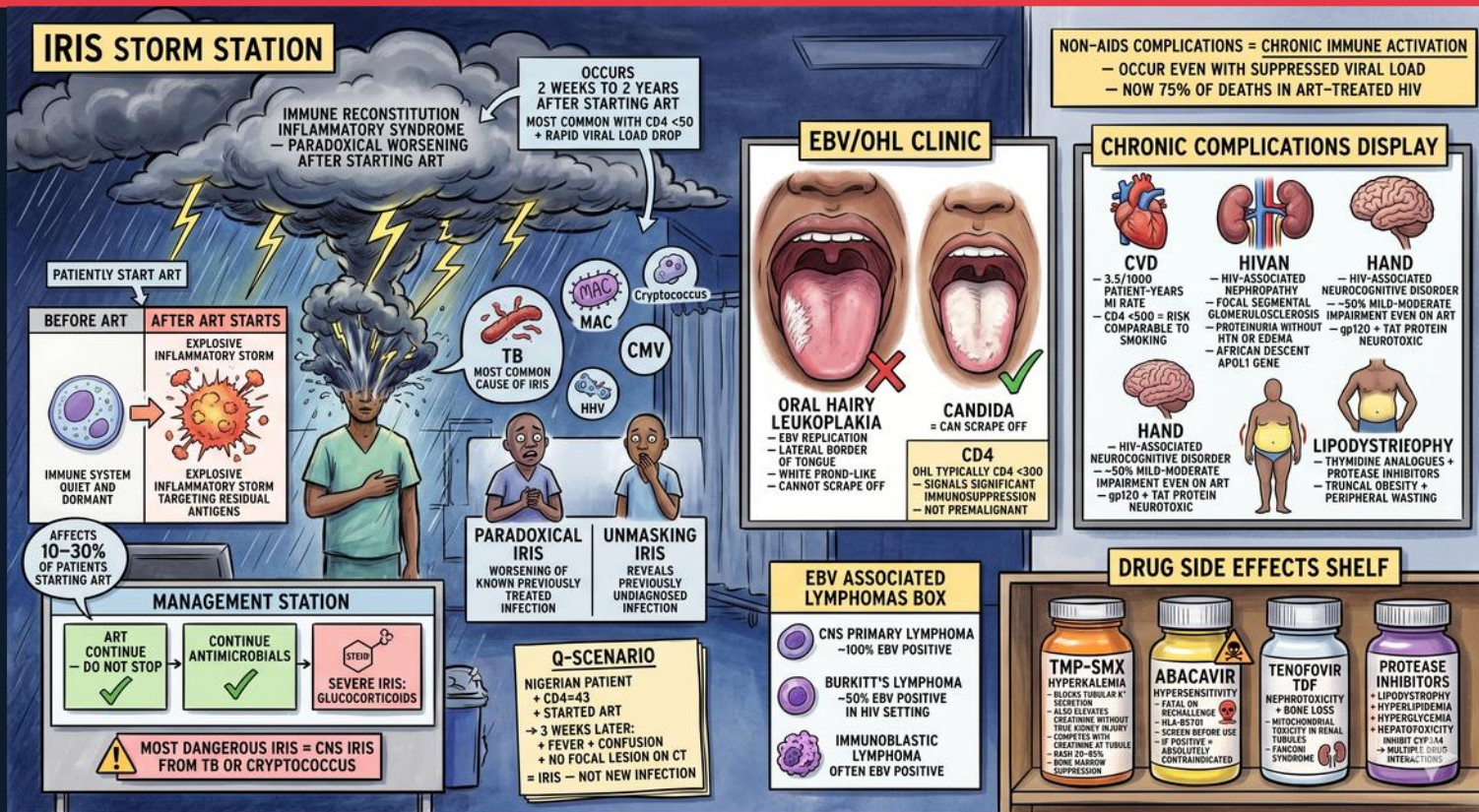
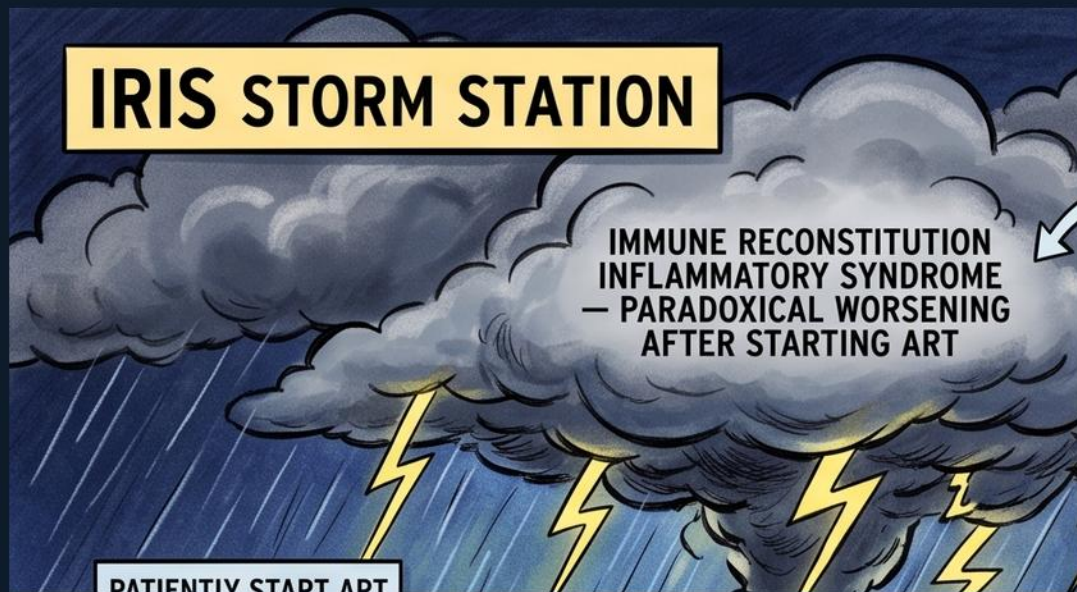


HIV — IRIS, EBV/OHL, Lymphomas & Drug Side Effects





IRIS storm station

WHAT YOU SEE

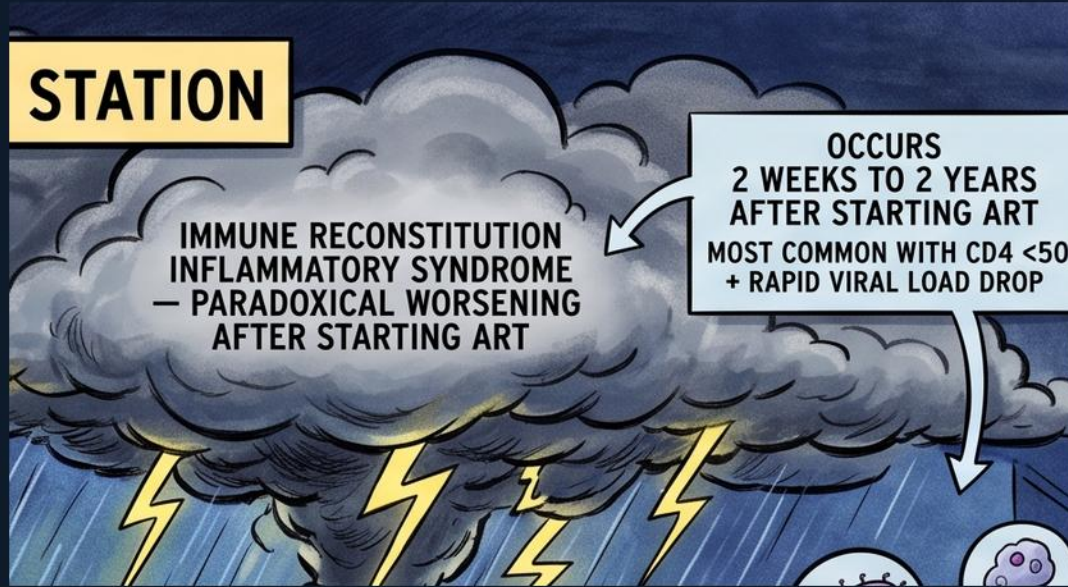
'IRIS Storm Station' lightning storm over patient starting ART

WHAT IT ENCODES

IRIS (immune reconstitution inflammatory syndrome) is a PARADOXICAL clinical worsening that occurs after starting ART: as the recovering immune system regains the ability to mount inflammation, it overreacts to antigens of a pre-existing (often subclinical or partially treated) opportunistic infection. The patient gets sicker even though the virus is being suppressed and CD4 is rising — an immune-driven, not infection-progression, deterioration.

THE BOARD TRAP

Wrong answer: 'a patient who worsens 3 weeks after starting ART has failed ART / has a resistant virus.' Discriminator: worsening WITH a falling viral load and rising CD4 is IRIS, not treatment failure. Check that VL is dropping — if it is, the deterioration is immune reconstitution, and ART is continued.



Timing 1wk-2yr

WHAT YOU SEE

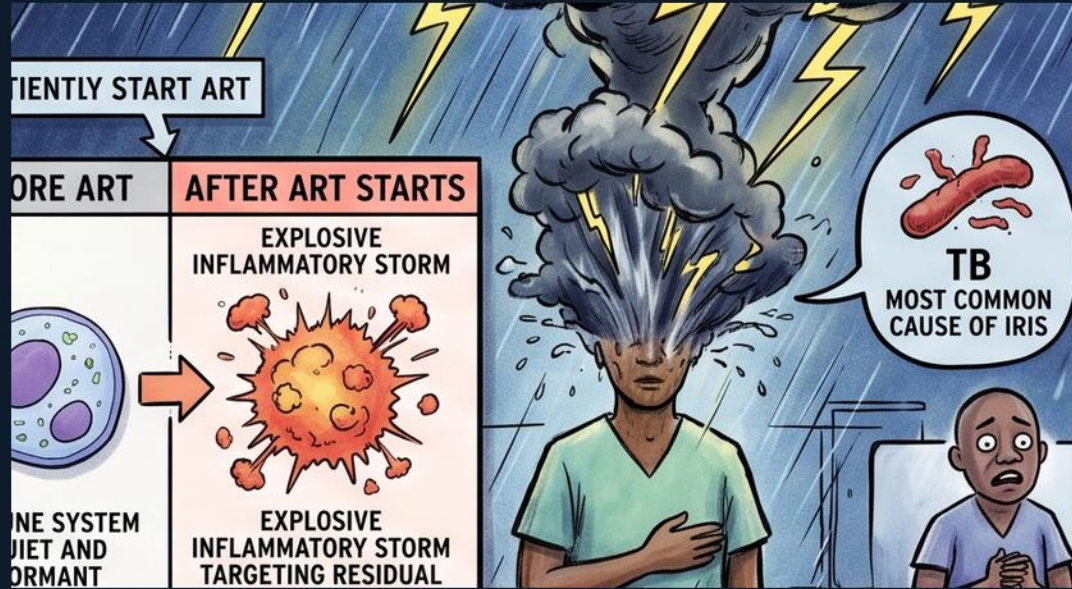
'Occurs ~2 weeks to 2 years after starting ART; most common with low CD4 + rapid VL drop'

WHAT IT ENCODES

IRIS typically appears weeks to a few months after ART initiation (range ~1 week to 2 years). The biggest risk factors are a very LOW baseline CD4 (especially <50), a high baseline viral load, a large/rapid drop in viral load, and a recently/concurrently diagnosed opportunistic infection. The lower the starting CD4 and the steeper the rebound, the higher the IRIS risk.

THE BOARD TRAP

Wrong answer: 'IRIS risk is highest in patients with high baseline CD4 counts.' Discriminator: it's the opposite — the lowest baseline CD4 (advanced immunosuppression) with the most dramatic VL fall carries the GREATEST IRIS risk, because there is more residual antigen and a bigger immune rebound.



Paradoxical IRIS

WHAT YOU SEE

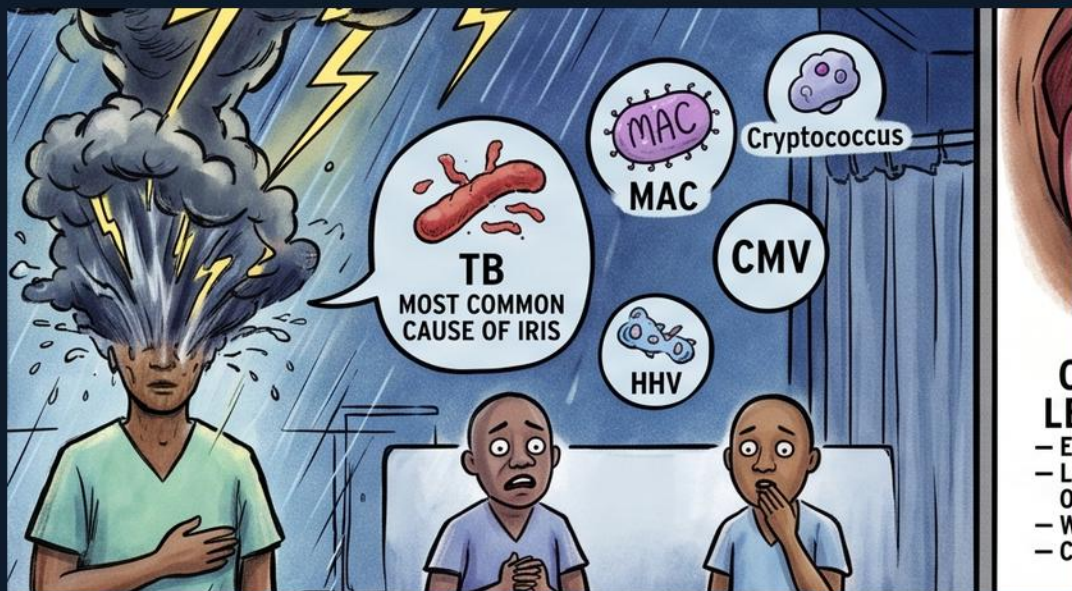
'Paradoxical IRIS — worsening of known/treated infection'

WHAT IT ENCODES

Paradoxical IRIS = clinical flare/worsening of an **ALREADY-DIAGNOSED** and treated opportunistic infection after ART-driven immune recovery (e.g., enlarging TB lymph nodes or new fevers in a patient already on TB therapy, or worsening cryptococcal meningitis). The pathogen is being treated; the deterioration is from the reconstituted inflammatory response to residual antigen.

THE BOARD TRAP

Wrong answer: 'enlarging lymph nodes in a TB patient after starting ART means the TB regimen is failing — escalate/change anti-TB drugs.' Discriminator: in a treatment-adherent patient with declining VL, this is paradoxical IRIS — continue both TB therapy and ART, add steroids if severe; don't assume drug resistance/failure.



Unmasking IRIS

WHAT YOU SEE

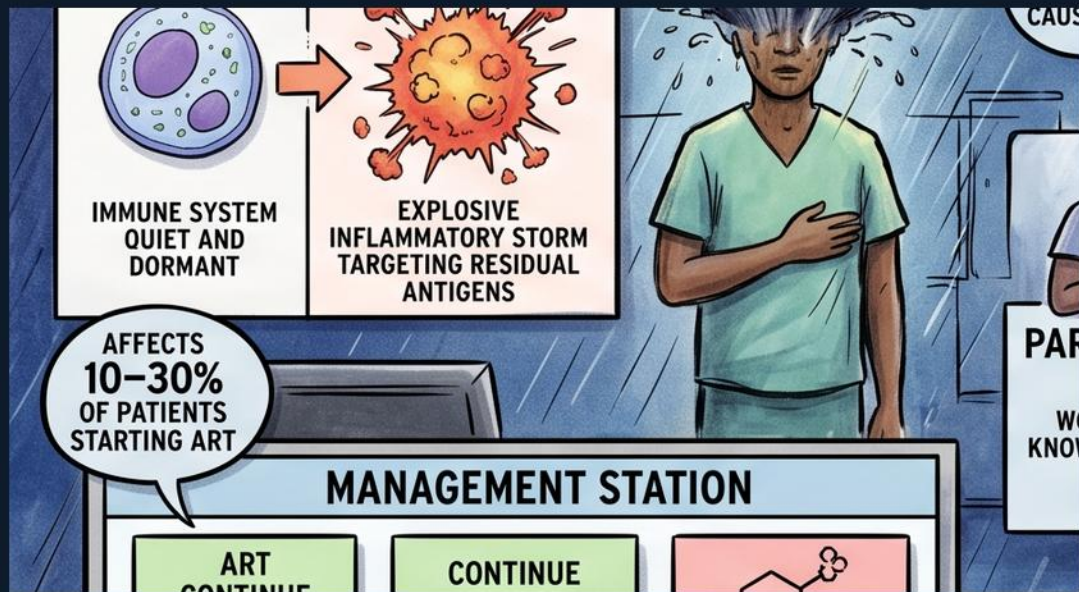
'Unmasking IRIS — reveals previously undiagnosed infection'

WHAT IT ENCODES

Unmasking IRIS = a previously latent/subclinical, undiagnosed infection becomes clinically apparent for the first time after ART because the recovering immune system suddenly reacts to it (e.g., new florid MAC lymphadenitis or cryptococcal disease appearing shortly after ART start in someone with no prior diagnosis). It 'unmasks' occult infection.

THE BOARD TRAP

Wrong answer: 'a new opportunistic infection appearing after ART proves the patient acquired it on ART / that ART is ineffective.' Discriminator: this is unmasking IRIS revealing a pre-existing occult infection, not a newly acquired one or ART failure — the diagnostic clue is the temporal link to immune recovery.



Continue ART (don't stop)

WHAT YOU SEE

'Management Station: ART continue — do NOT stop; treat OI'

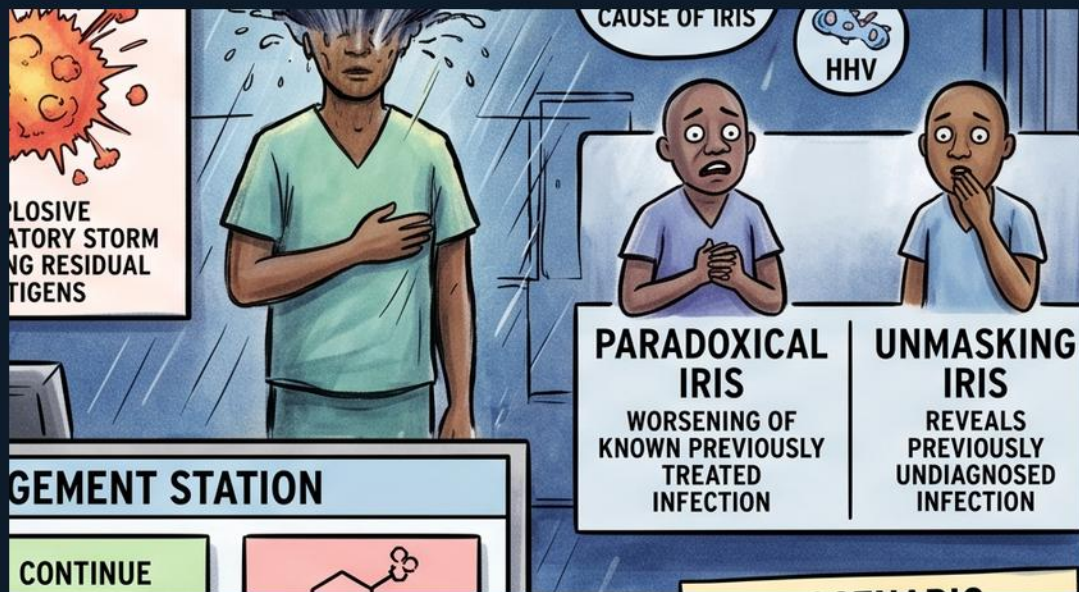
WHAT IT ENCODES

Core management of IRIS: CONTINUE ART and treat/optimize therapy for the underlying opportunistic infection. Most IRIS is self-limited. Stopping ART risks viral rebound, resistance, and further immunosuppression, and is reserved only for truly life-threatening situations that cannot be controlled otherwise.

THE BOARD TRAP

Wrong answer: 'stop ART until the IRIS reaction resolves.'

Discriminator: for the great majority of IRIS, you continue ART and treat the OI ($\pm$ steroids if severe). Discontinuing ART is the wrong move and is reserved for rare life-threatening, uncontrollable cases.



Severe IRIS → steroids

WHAT YOU SEE

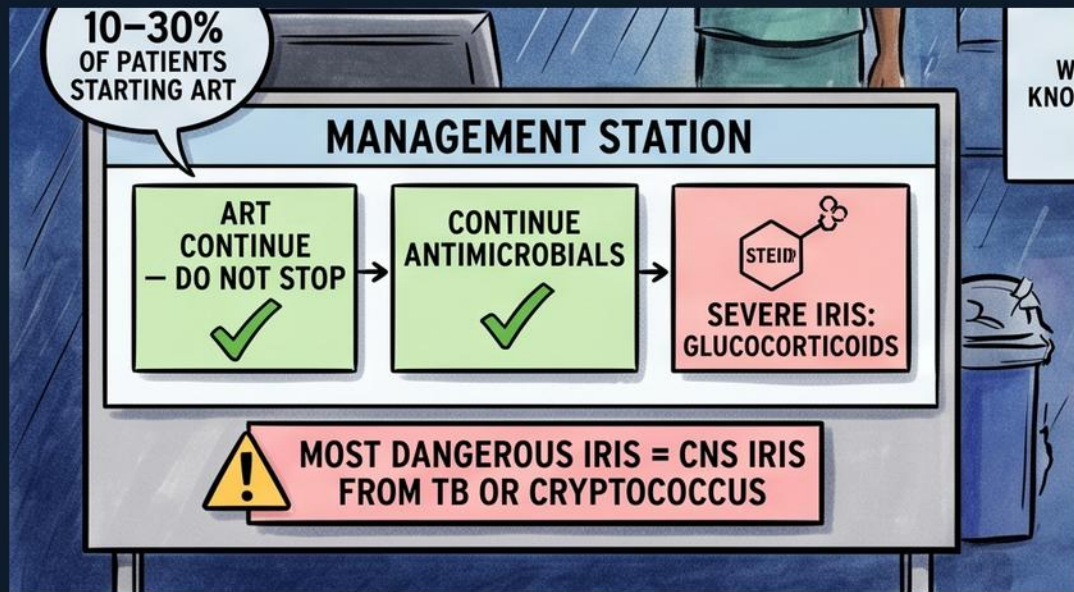
'Severe IRIS: glucocorticoids'

WHAT IT ENCODES

Severe or organ-threatening IRIS (especially CNS involvement, severe respiratory compromise) is treated with systemic CORTICOSTEROIDS to dampen the excessive inflammatory response while ART is continued and the OI is treated. Steroids reduce inflammation/edema (e.g., paradoxical TB-IRIS) without needing to interrupt antiretrovirals.

THE BOARD TRAP

Wrong answer: 'IRIS always requires high-dose steroids.'
Discriminator: most IRIS is mild and self-limited needing only supportive care + continued ART/OI treatment; steroids are reserved for SEVERE/organ-threatening disease (notably CNS or respiratory).



Dangerous CNS IRIS

WHAT YOU SEE

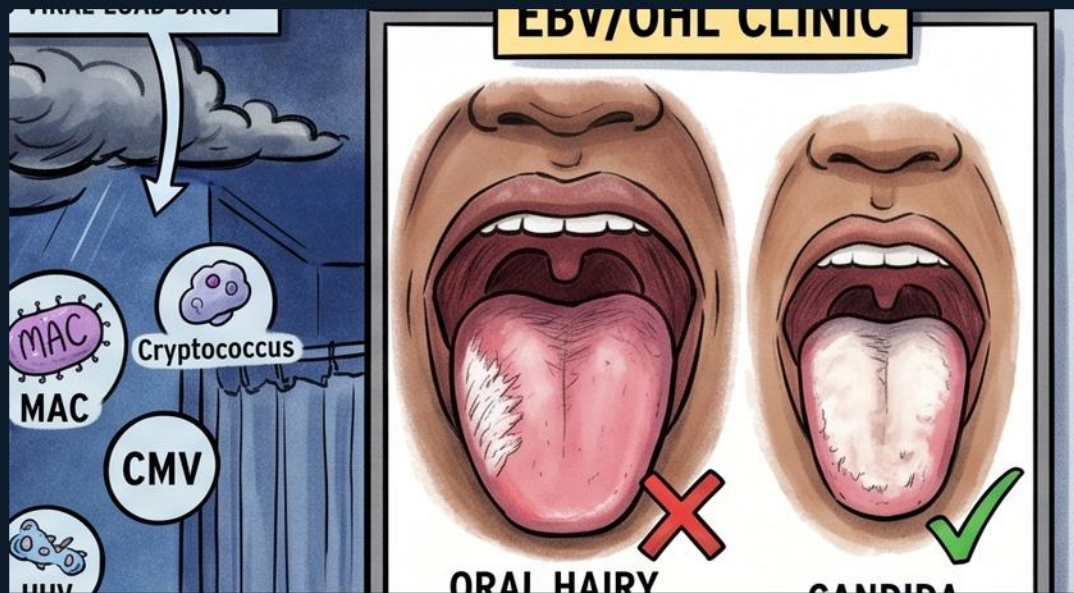
'Most dangerous = CNS IRIS from TB or cryptococcus'

WHAT IT ENCODES

CNS IRIS — particularly from tuberculous meningitis and cryptococcal meningitis — is the most dangerous form because inflammation in the closed cranial space causes raised ICP, edema, and herniation risk. This is precisely why ART is DELAYED in cryptococcal meningitis (start anti-fungal induction first, then begin ART after ~4–6 weeks) to avoid catastrophic CNS IRIS.

THE BOARD TRAP

Wrong answer: 'start ART immediately in a patient newly diagnosed with cryptococcal meningitis.' Discriminator: in cryptococcal (and to some extent TB) meningitis, ART should be DEFERRED a few weeks after antifungal/anti-TB therapy because immediate ART increases lethal CNS IRIS — a key exception to the 'treat everyone immediately' rule.



Oral hairy leukoplakia

WHAT YOU SEE

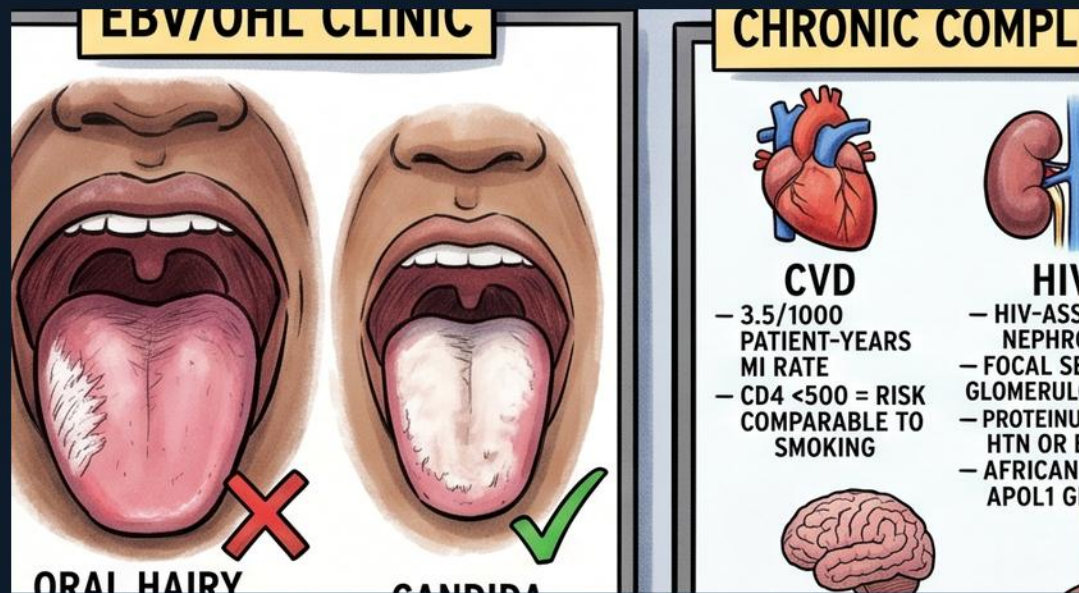
'EBV/OHL Clinic': tongue with white plaque on lateral border, cannot scrape off

WHAT IT ENCODES

Oral hairy leukoplakia (OHL) is an EBV-driven white, corrugated/'hairy' plaque on the LATERAL borders of the tongue that does NOT scrape off. It is a marker of immunosuppression (HIV or other), is benign and NON-premalignant, and often needs no specific treatment (responds to ART; antivirals only if symptomatic). It is not caused by Candida.

THE BOARD TRAP

Wrong answer: 'a non-scrapeable white tongue plaque is leukoplakia that is premalignant and needs biopsy/excision.'
 Discriminator: OHL is EBV-related, benign, and NON-premalignant — distinct from true (dysplastic) oral leukoplakia. It does not scrape off (unlike candida) and resolves with ART; aggressive intervention is unnecessary.



Candida (scrapes off)

WHAT YOU SEE

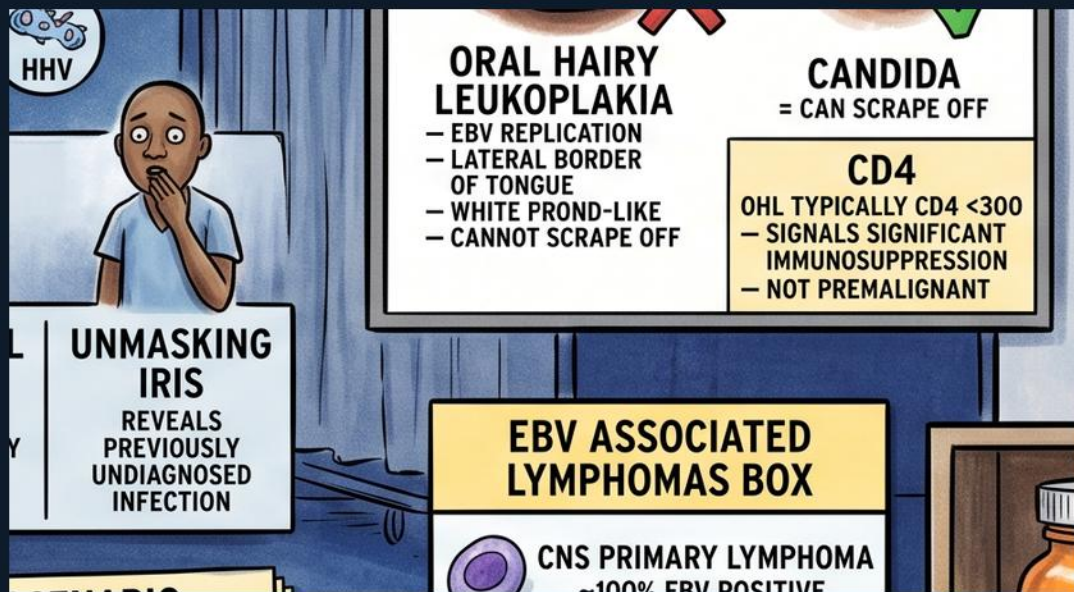
Adjacent tongue 'Candida — can scrape off', CD4 <200

WHAT IT ENCODES

Oral candidiasis (thrush): creamy white plaques that DO scrape off leaving an erythematous, sometimes bleeding base; KOH/smear shows pseudohyphae. It signals CD4 <200 (or other immunosuppression) and warrants checking CD4 and starting PCP prophylaxis if indicated. Treat with topical (nystatin/clotrimazole) or oral fluconazole for extensive/esophageal disease.

THE BOARD TRAP

Wrong answer: 'a white oral plaque that scrapes off easily is oral hairy leukoplakia.' Discriminator: SCRAPES OFF = Candida (thrush); does NOT scrape off and on lateral tongue = OHL (EBV). The scrape test is the classic bedside differentiator, and thrush flags likely CD4 <200.



EBV lymphoma box

WHAT YOU SEE

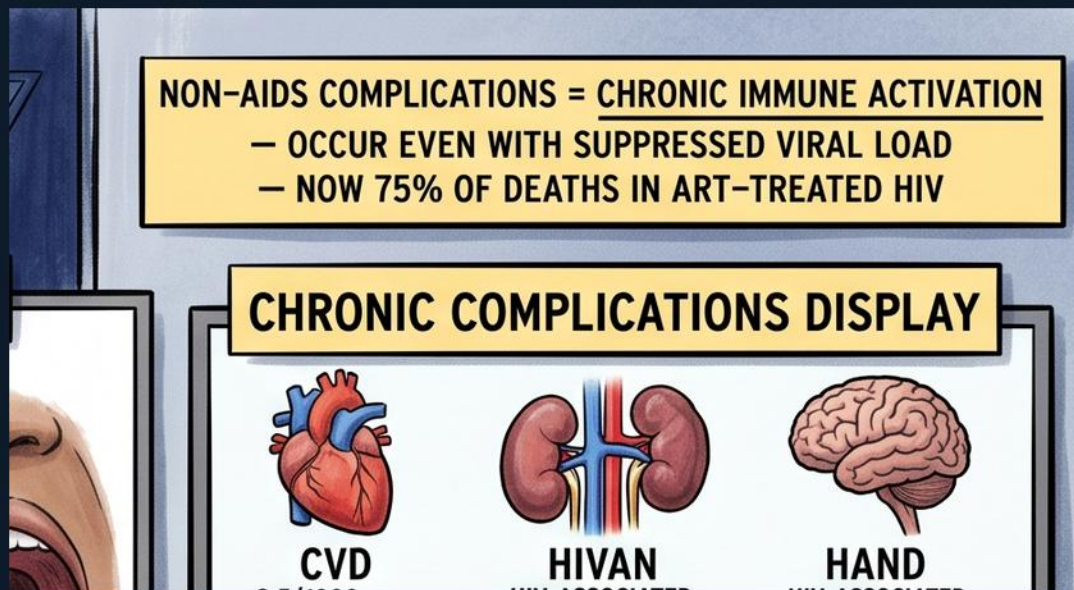
'EBV-Associated Lymphoma Box': CNS primary lymphoma, Burkitt, immunoblastic

WHAT IT ENCODES

EBV drives several HIV-associated lymphomas: primary CNS lymphoma (nearly 100% EBV-positive, typically CD4 <50), Burkitt lymphoma (EBV-associated, 't(8;14)' c-MYC, 'starry-sky'), and immunoblastic/diffuse large B-cell lymphoma. CSF EBV PCR positivity supports primary CNS lymphoma in the workup of a brain mass.

THE BOARD TRAP

Wrong answer: 'a solitary enhancing brain lesion in advanced HIV is most likely toxoplasmosis — treat empirically and never biopsy.' Discriminator: a SINGLE periventricular lesion with positive CSF EBV PCR (and/or thallium/PET uptake) favors primary CNS LYMPHOMA over toxoplasmosis (which is usually multiple). Lack of response to empiric anti-toxo therapy → biopsy for lymphoma.



Chronic complications

WHAT YOU SEE

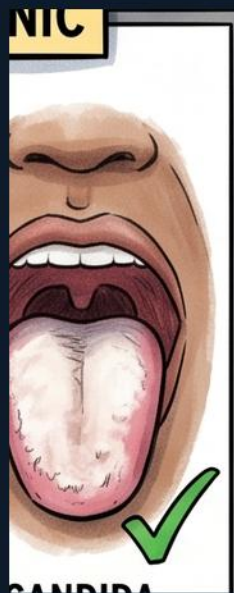
'Chronic Complications Display': CVD, HIVAN, neurocognitive, lipodystrophy

WHAT IT ENCODES

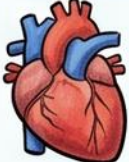
Even with virologic suppression, persistent chronic immune activation/inflammation drives non-AIDS morbidity: accelerated cardiovascular disease, HIV-associated nephropathy (HIVAN), HIV-associated neurocognitive disorder (HAND), and metabolic complications/lipodystrophy. In the ART era, the MAJORITY (>75%) of deaths in treated HIV are from these non-AIDS chronic conditions rather than opportunistic infections.

THE BOARD TRAP

Wrong answer: 'most deaths in well-controlled HIV patients today are from opportunistic infections.' Discriminator: with effective ART, the leading causes of death shifted to NON-AIDS chronic complications (cardiovascular disease, renal, hepatic, malignancy) — OIs are now the minority of deaths.



CHRONIC COMPLICATIONS DISPLAY

CVD	HIVAN	HAND
 — 3.5/1000 PATIENT-YEARS MI RATE — CD4 <500 = RISK COMPARABLE TO SMOKING	 — HIV-ASSOCIATED NEPHROPATHY — FOCAL SEGMENTAL GLOMERULOSCLEROSIS — PROTEINURIA WITHOUT HTN OR EDEMA — AFRICAN DESCENT APOL1 GENE	 — HIV-ASSOCIATED NEUROCOGNITIVE DISORDER — ~50% MILD-MODERATE IMPAIRMENT EVEN ON ART — gp120 + TAT PROTEIN NEUROTOXIC

HIVAN

WHAT YOU SEE

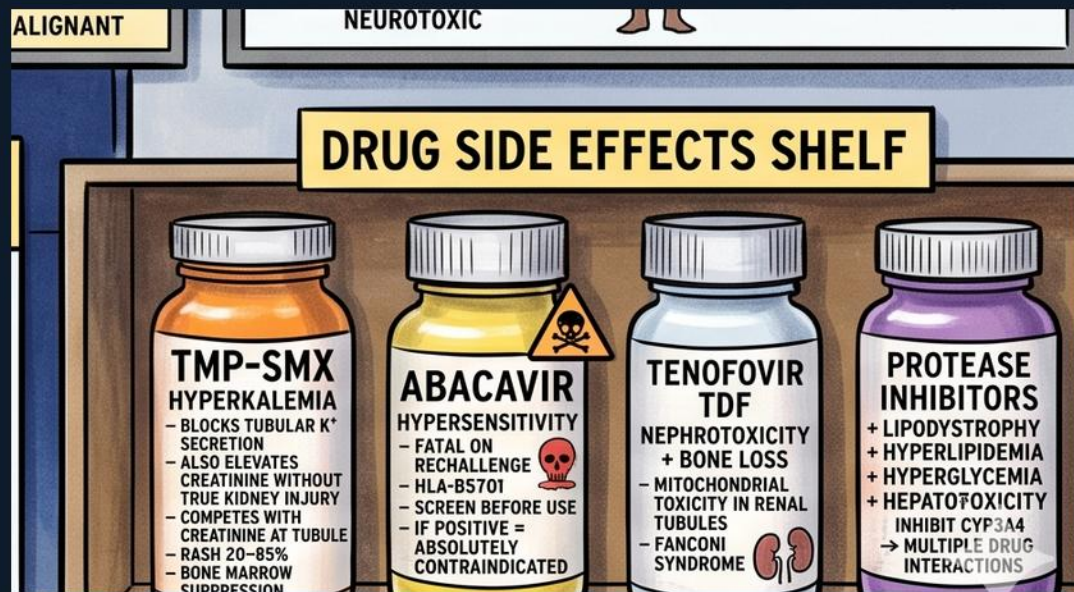
'HIVAN' kidney, FSGS / collapsing glomerulopathy

WHAT IT ENCODES

HIV-associated nephropathy (HIVAN) is classically **COLLAPSING** focal segmental glomerulosclerosis with heavy (often nephrotic-range) proteinuria, rapidly declining renal function, and large echogenic kidneys on ultrasound (kidneys not shrunk despite renal failure). It is strongly associated with Black patients (APOL1 risk alleles) and uncontrolled HIV. The primary treatment is ART ($\pm$ ACE inhibitor/ARB and sometimes steroids).

THE BOARD TRAP

Wrong answer: 'treat HIVAN primarily with immunosuppressants/cyclophosphamide.' Discriminator: the cornerstone of HIVAN treatment is ART (viral suppression), plus RAAS blockade for proteinuria — not aggressive cytotoxic immunosuppression. Note the paradox of **LARGE** echogenic kidneys despite renal failure.



TMP-SMX side effect

WHAT YOU SEE

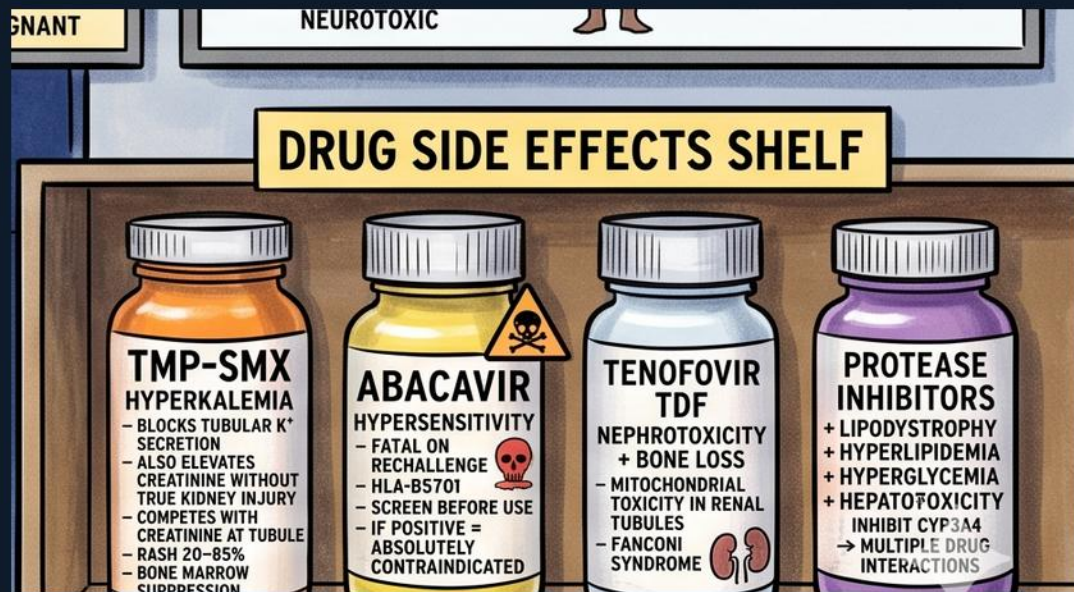
Drug Side Effects Shelf: TMP-SMX → hyperkalemia, rash, marrow suppression

WHAT IT ENCODES

Trimethoprim-sulfamethoxazole (used for PCP/toxo prophylaxis and treatment) adverse effects: **HYPERKALEMIA** (trimethoprim blocks the distal tubule ENaC channel, amiloride-like), a rise in creatinine (blocks tubular creatinine secretion without true GFR loss), rash/photosensitivity up to Stevens-Johnson syndrome, bone marrow suppression/cytenopias, and hemolysis in G6PD deficiency.

THE BOARD TRAP

Wrong answer: 'a creatinine bump on TMP-SMX means acute kidney injury — stop the drug.' Discriminator: trimethoprim blocks tubular **CREATININE** secretion, raising serum creatinine **WITHOUT** reducing true GFR (pseudo-AKI). Combined with its hyperkalemia effect, this is expected and not necessarily nephrotoxicity. (True hyperkalemia, though, can be dangerous and needs monitoring.)



Abacavir HLA-B*5701

WHAT YOU SEE

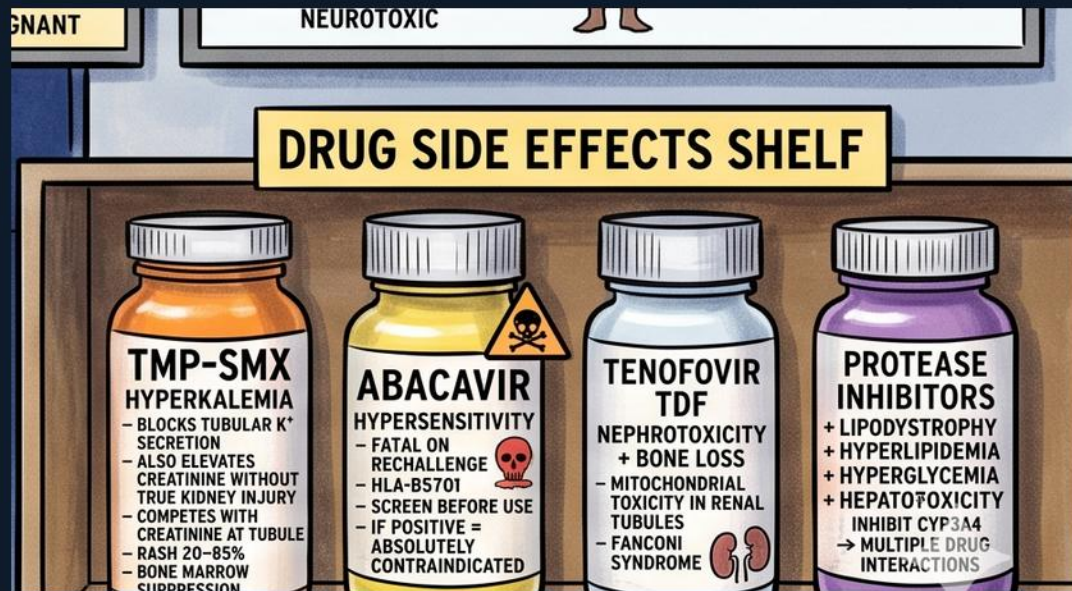
Abacavir bottle, hypersensitivity reaction

WHAT IT ENCODES

Abacavir (NRTI) can cause a potentially fatal hypersensitivity reaction (fever, rash, GI and constitutional/respiratory symptoms, worsening on rechallenge) tightly linked to the HLA-B*5701 allele. Screen for HLA-B*5701 BEFORE prescribing; if positive, abacavir is contraindicated. Abacavir has also been associated in some studies with increased cardiovascular risk.

THE BOARD TRAP

Wrong answer: 'rechallenge with abacavir after the hypersensitivity symptoms resolve.' Discriminator: NEVER rechallenge abacavir after a hypersensitivity reaction — re-exposure can cause fatal reactions. And the drug must not be started at all without prior HLA-B*5701 testing (positive = absolute contraindication).



Tenofovir renal/bone

WHAT YOU SEE

Tenofovir bottle, renal + bone toxicity

WHAT IT ENCODES

Tenofovir disoproxil fumarate (TDF) causes nephrotoxicity via PROXIMAL renal tubulopathy (Fanconi syndrome: phosphaturia, glucosuria with normal glucose, proteinuria, acidosis) and reduced bone mineral density. Tenofovir alafenamide (TAF) delivers the active drug with much lower plasma tenofovir levels and therefore significantly LESS renal and bone toxicity, preferred in patients with renal/bone risk.

THE BOARD TRAP

Wrong answer: 'TDF and TAF have equivalent renal and bone safety.' Discriminator: TAF causes markedly LESS renal (proximal tubulopathy/Fanconi) and bone toxicity than TDF and is preferred when kidney or bone disease is a concern (TDF, however, has a more favorable lipid profile).

Summary — every symbol

1. IRIS storm station

IRIS (immune reconstitution inflammatory syndrome) is a PARADOXICAL clinical worsening that occurs after starting ART: as the recovering immune system regains the ability to mount inflammation, it overreacts to antigens of a pre-existing (often subclinical or partially treated) opportunistic infection

3. Paradoxical IRIS

Paradoxical IRIS = clinical flare/worsening of an ALREADY-DIAGNOSED and treated opportunistic infection after ART-driven immune recovery (e.g., enlarging TB lymph nodes or new fevers in a patient already on TB therapy, or worsening cryptococcal meningitis)

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Core management of IRIS: CONTINUE ART and treat/optimize therapy for the underlying opportunistic infection

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IRIS typically appears weeks to a few months after ART initiation (range ~1 week to 2 years)

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